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SHAP-Driven Machine Learning Approaches for LST Modeling Using Surface Indices in Chuadanga

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Highlights

- The different patterns in the urban intensity (NDBI -0.26 to -0.02), vegetation density (NDVI 0.14-0.42), waterbodies (MNDWI -0.24 -0.05), surface temperature (LST-31.27 to 39.25 °C) and thermal stress levels (UTFVI -0.11 to 0.49) were observed in the spatial indices.
- Correlations between NDBI-LST and NDVI-LST (0.68 and -0.54) and NDBI and NDVI (-0.89) were considered to be strong.
- It was also found that SVM had the highest performance where $r = 0.63$, $R^2 = 0.393$, RMSE = 1.71 °C, MAE = 1.02 °C and the highest AUC = 0.8115 as compared to the other three models, yet all the models exhibited a strong discriminatory ability.
- SHapley Additive exPlanations analysis reveals that NDBI was the most significant characteristic (0.942) in the machine learning model.

Abstract

The rapid urbanization and climate change has increased the surface heating of urban peri urban areas in Bangladesh. This paper analyzes the land surface temperature within Chuadanga District using Landsat 8 imageries of April 2024 and discusses the impact of vegetation, built-up surfaces and water bodies based on NDVI, NDBI and MNDWI measures. The spatial indices were as NDBI -0.26 to 0.02, NDVI 0.14 to 0.42, MNDWI -0.24 to -0.05, and LST 31.27 to 39.25 °C. There existed strong correlations among NDBI NDVI (-0.89), NDBI NDVI LST (0.68), and NDVI LST (-0.54), which shows that urban expansion increases the surface temperature, and vegetation alleviates the temperature. To predict LST, four machine learning models were used, which are the Random Forest (RF), the Support Vector Machine (SVM), the Extreme Gradient Boosting (XGBoost), and the Artificial Neural Network (ANN). The best model in terms of $r = 0.63$, $R^2 = 0.393$, RMSE 1.71 °C and MAE 1.02 °C and the largest AUC of 0.8115 was Support Vector Machine. Analysis using SHapley Additive exPlanations (SHAP) found NDBI to be the strongest effectively causing surface temperature, NDVI, and MNDWI came in second and third. The Index of Urban Thermal Field Variance (-0.11 to 0.49) indicated the presence of high thermal stress that was localized in the urban cores and industrial areas that revealed the role of urbanization in amplifying the Urban Heat Island effect and the buffering effect of vegetation and water bodies. The paper has shown that remote sensing, spectral indices, and machine learning can be used to offer a solid framework of LST evaluation as well as provide arguments of evidence-based urban climate adaptation, such as LST-based zoning, nature-based mitigation plans, and policy-making in rapidly developed cities.

Keywords: LST, Machine Learning, RF, SVM, ANN, XGBoost, SHAP, Chuadanga, Bangladesh.

Abbreviations

ANN	Artificial Neural Networks
LST	Land Surface Temperature
MAE	Mean Absolute Error
ML	Machine Learning
MNDWI	Modified Normalized Difference Water Index
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
r	Coefficient
R²	Coefficient of Determination
RF	Random Forest
RMSE	Root Mean Squared Error
RMSLE	Root Mean Squared Logarithmic Error
ROC	Receiver Operating Characteristic
SD(σ)	Standard Deviation
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SUHI	Surface Urban Heat Islands
SVM	Support Vector Machine
SVR	Support Vector Regression
UHI	Urban Heat Island
UTFVI	Urban Thermal Field Variance Index
XGBoost	eXtreme Gradient Boosting

1 Introduction

Climate change amplifies the frequency, intensity, and duration of extreme heat events (Perkins-Kirkpatrick & Lewis, 2020; F. Zha et al., 2024), mostly in urban areas where 68% of the global population is expected to live by 2050 (Nations et al., 2018). By the late 20th century, the transformation of landscape has been substantial with settlements covering 1 to 6 percentage of earth's surface (Rasool et al., 2021; Uisso & Tanrıvermiş, 2021). Moreover, forest and grassland areas have decreased by 19% and 8%, respectively, over the past three centuries (Song, 2018; Steven et al., 2011). Urban expansion is characterized by growth or shrinkage which may transform natural vegetation or water body into impervious surface (Jiang et al., 2020; Viana et al., 2019). Urban areas are generally 1 °C to 5°C warmer than rural regions, attributable to diminished vegetation and increased impervious surfaces (Abd-Elmabod et al., 2022; Farid et al., 2022; Krasniqi & Rapuca, 2024; Yuan et al., 2019; X. Zhou & Chen, 2018). Moreover, urbanization exacerbates the UHI effect by 2 to 8°C (Choe & Yom, 2020; Rashid et al., 2022).

The intensity of the urban heat island (UHI) effect indicates a spatial variation across countries, primarily due to differences in morphological and surface characteristics (Masson et al., 2020; Mentaschi et al., 2022). Land surface temperature (LST) represents the radiative energy exchange between the Earth's surface and the atmosphere (Avdan & Jovanovska, 2016; Z. L. Li et al., 2013; Mustafa et al., 2020; Tomlinson et al., 2011). The normalized difference vegetation index (NDVI) is commonly used to estimate LST and analyze vegetation patterns in urban areas, as lower vegetation cover is generally associated with higher LST values. While correlations between vegetation and surface temperature have been extensively explored (Ferrelli et al., 2018). Respectively, the modified normalized difference water index (MNDWI) and normalized difference built up index (NDBI) are frequently employed to analyze LST patterns in relation to water bodies and built-up areas (Ghosh et al., 2019). A negative correlation was observed between vegetation or waterbodies and LST (Ogashawara & Bastos, 2012), whereas a positive relationship was found between NDBI and LST(X. L. Chen et al., 2006; L. Li

et al., 2020). Vegetation contributes to the natural cooling effect by absorbing solar radiation (Fleck et al., 2022; Marando et al., 2022; Robbiati et al., 2022).

Macarof & Stătescu (2017) found a strong linear relationship between LST and NDBI, indicating that as built-up areas increase, so does the LST. A positive correlation was observed between LST and NDBI, while the influence of NDVI and MNDWI on LST was also assessed using various spectral indices. In urbanized regions, higher NDBI values correspond to elevated temperatures, suggesting that urbanization contributes to the urban heat island effect (Cetin et al., 2024). Meng et al. (2022) and Xiang et al. (2023) also identified that NDBI was positively correlated with LST across all seasons, strongest in summer and weakest in winter, while NDVI and MNDWI showed considerable seasonal variation, with significant differences in correlation strength observed in winter compared to spring, summer, and autumn. Additionally, LST exhibits a strong negative correlation with NDVI, indicating that areas with higher vegetation density tend to have lower temperatures (Guha et al., 2020), which demonstrate that increased vegetation cover contributes to a reduction in LST, emphasizing the cooling effect of green spaces (Sobremonte-Maglipon et al., 2024). In Egypt, it was demonstrated that LST exhibited a positive correlation with NDBI and negative correlations with NDVI and NDWI (Morsy & Hadi, 2022). Polynomial regression analysis was used, which yielded an R^2 value of 0.628 for NDBI, while NDVI and NDWI had values of 0.341 and 0.305, respectively. Sameh et al. (2022) demonstrated a regression analysis, which showed that NDBI and NDBSI significantly contribute to increases in LST, explaining approximately 80–85% of its variation, whereas NDVI accounts for a 63–68% decrease in LST, indicating a negative correlation between LST and vegetation cover.

Liang et al. (2023) utilized NDVI, NDBI, and DEM as covariates in the GNNWR model to establish a regression relationship with LST, although they did not include MNDWI as a factor in the downscaling process. This method integrates NDVI, NDBI, and other spatial data to achieve high-resolution LST downscaling, demonstrating R^2 0.974 and RMSE of 0.896°C, thereby indicating its effectiveness in capturing spatial heterogeneity. Convolutional neural networks (CNN) and long short-term memory (LSTM) models can also be adapted to forecast NDVI, which in turn influences LST predictions (Gao et al., 2023). They utilized TSD-CNN-LSTM model, which showed superior performance in NDVI forecasting, crucial for understanding vegetation's impact on LST. On the other hand, Xia et al. (2018) discusses using the regression methods to predict high-resolution LST based on visible and near-infrared (VNIR) and thermal infrared (TIR) data, though they did not specifically address the relationships between LST and NDBI, NDVI, or MNDWI. Several studies have shown a positive relationship between NDBI and LST, especially during winter (Guha et al., 2020). SHAP analysis revealed both positive and negative NDBI effects, built-up areas tended to increase LST, while cooling elements such as vegetation and rooftop gardens contributed to its reduction. NDVI exhibited a stronger influence during winter; in areas with sparse vegetation, higher NDVI values could contribute to increased LST (Julien et al., 2006; Kaufmann et al., 2003). This fluctuation has been documented since the 19th century (Alexander, 2020; Howard, 2012; Oke, 1982), highlighting the urgent need for urban policies aligned with SDG 11.

Spatial interconnections (S. Li et al., 2010) and localized variations within surface indicators for forecasting LST can present difficulties in being captured efficiently and precisely (Lu et al., 2014; Wang et al., 2017; Zhang et al., 2005). Recently machine learning has found growing application, in GIS and remote sensing research, which operates at the crossroads of computer science, intelligence and data science. It highlights the use of data and algorithms to support learning and improve accuracy and efficiency progressively (IBM, 2025). Traditional machine learning approaches commonly employ least squares (OLS) and multiple linear regression (MLR) to forecast upcoming trends or results (Q. Li & Zheng, 2023). Hovorushchenko & Alekseiiko, (2024) noted the increasing adoption of machine learning in sensing and described how conventional techniques have difficulty identifying nonlinear patterns, in environmental datasets. In contrast, machine learning models learn from predefined assumptions which enhance the predictive accuracy and generalization capabilities (Hovorushchenko et al., 2024).

Chudanga district recorded a highest temperature of 42.2 °C on April 21, 2024, reflecting the ongoing pattern of extreme heat events in that region (M. A. Rahman et al., 2024). This study aims to predict land surface temperature and examine its association with key surface indices and urban thermal conditions. Four advanced machine

learning models—Random Forest (RF), Extreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), and Artificial Neural Network (ANN)—are utilized to predict surface temperature based on three primary indices: the Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), and Modified Normalized Difference Water Index (MNDWI). The SHapley Additive exPlanations (SHAP) framework is employed to interpret the model's predictions and determine the relative contribution of each variable to temperature variation. Additionally, the Urban Thermal Field Variance Index (UTFVI) is applied to delineate risk zones and assess spatial patterns of thermal stress for policy concerns and implementation.

2 Methodology

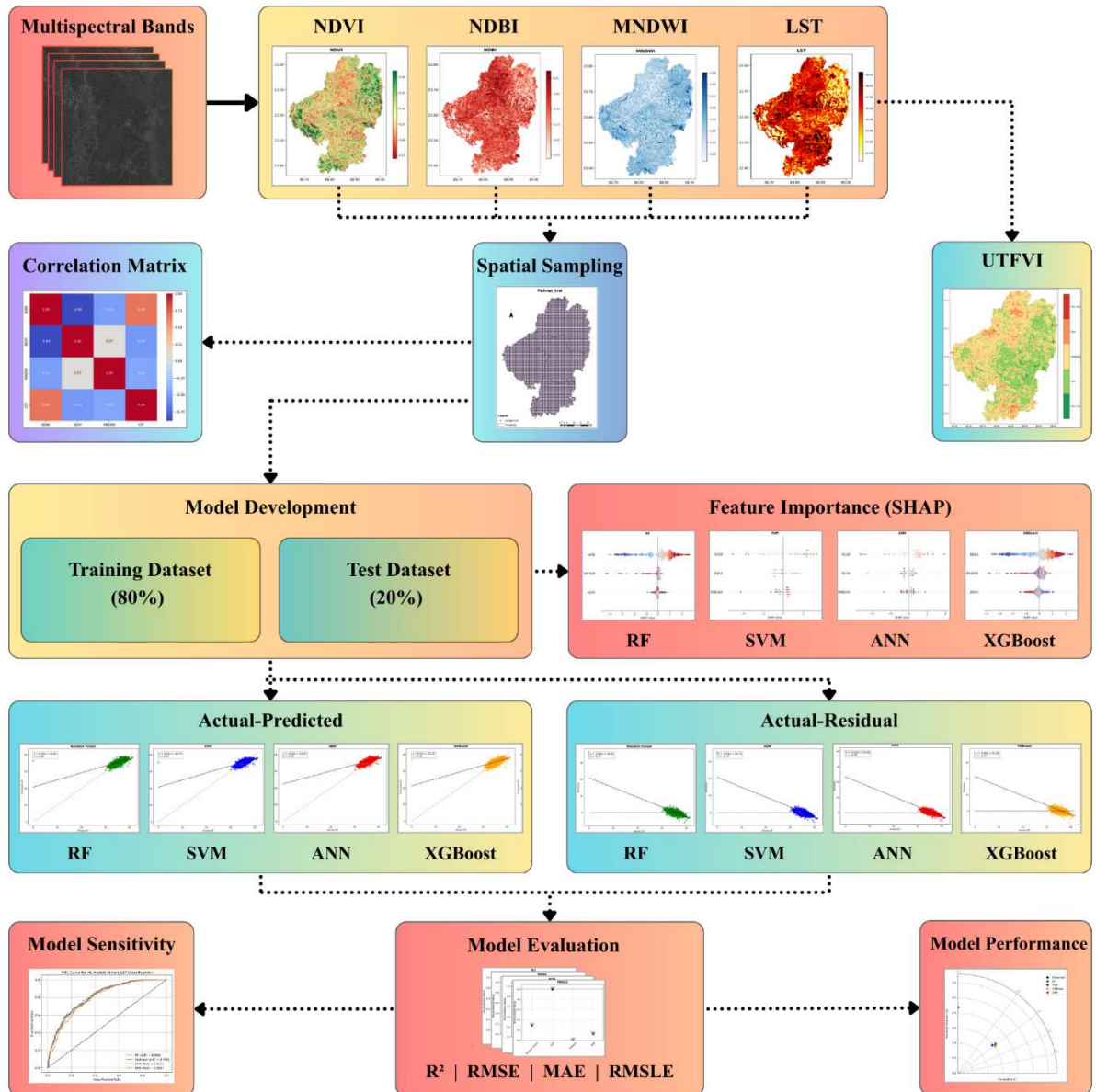


Figure 1: Methodological framework of the study.

2.1 Study Area

This research was undertaken in the district of Chuadanga, located in the southwestern region of Bangladesh, geographically positioned between the latitudes 23°22' and 23°50' north and longitudes of 88°39' and 89°00' east. The area encompasses 1,157.42 square kilometers (446.88 square miles) and is subdivided into four subdistricts: Chuadanga Sadar, Damurhuda, Alamdanga, and Jibannagar. It is bordered to the north by Kushtia and Meherpur districts, to the south and east by Jhenaidah district, and to the west by Meherpur district and the Indian state of West Bengal (Halim et al., 2008; Islam et al., 2023).

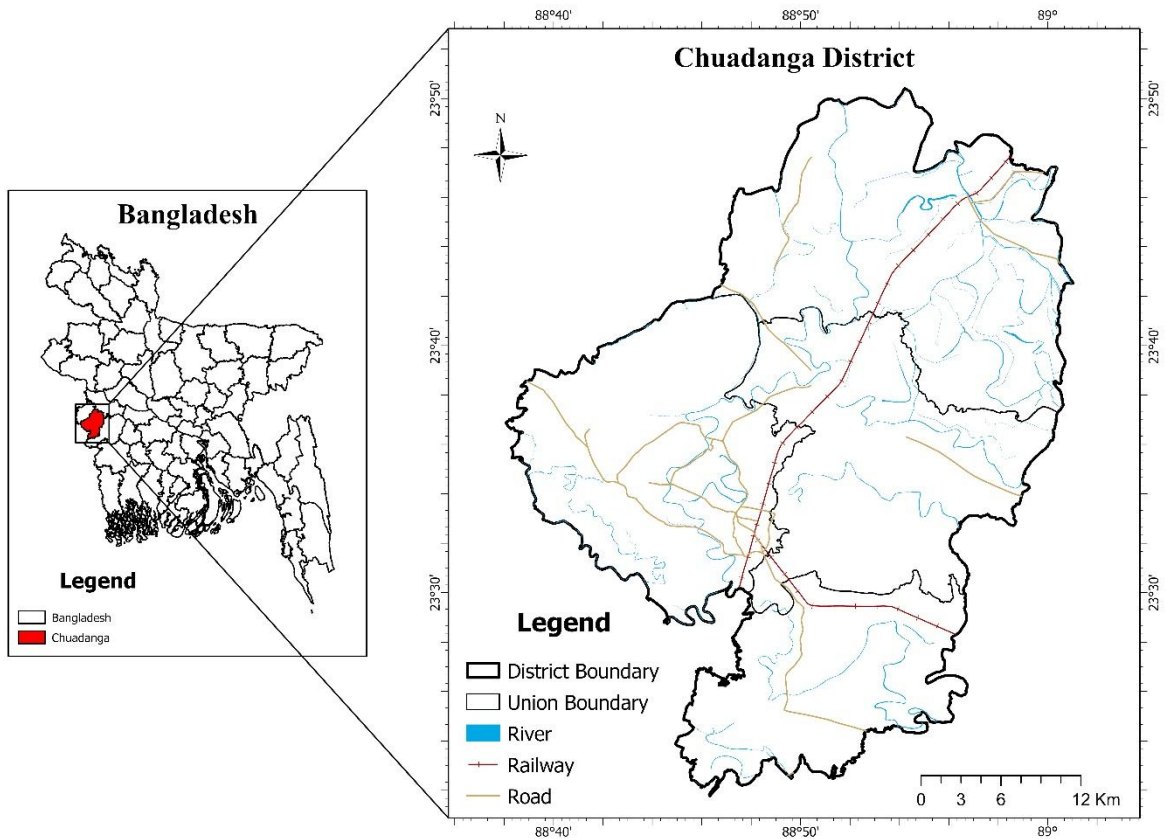


Figure 2: Study area map [Prepared by the authors].

2.2 Tools

The study utilized a variety of software tools, including Microsoft Excel for data entry and management, SPSS for statistical analysis, Google Earth Engine, ArcGIS, and QGIS for spatial analysis, and Python for advanced data processing and visualization.

2.3 Data

Landsat 8 Collection 2 Level 2 Surface Reflectance (SR) imagery, acquired on 1 April 2024 (Path/Row: 138/044), with cloud cover less than 3%, was selected to ensure the accurate seasonal representation of land surface conditions. Preprocessing employed the Level 2 product's integrated quality assurance features, including the QA_PIXEL band produced by the CF Mask algorithm, which automatically identifies and masks clouds, shadows, and atmospheric artifacts—thus eliminating the need for manual masking (Foga et al., 2017). Spectral reflectance bands—SR_B3 (Green), SR_B4 (Red), SR_B5 (Near-Infrared), and SR_B6 (Shortwave Infrared)—were extracted for spectral index computation, while the thermal band ST_B10 underwent atmospheric correction to derive Land Surface Temperature (LST).

Table I:P Data Acquisition for the study

Parameter	Details
Scene ID	LC08_L2SP_138044_20240401_20240410_02_T1
Acquisition Date	01 April 2024
Path/Row	138/044
Cloud Cover	0% to <5%
Processing Level	Level-2 Surface Reflectance (SR)
Bands	SR_B3 (Green), SR_B4 (Red), SR_B5 (NIR), SR_B6 (SWIR1), ST_B10 (Thermal Infrared)

2.4 Methods

2.4.1 Normalized Difference Vegetation Index (NDVI)

NDVI is one of the most widely used vegetation metrics, capturing greenness and chlorophyll activity. Where *NIR* and *RED* correspond to near-infrared and red reflectance bands. NDVI values range from -1 to $+1$, with higher values indicating dense vegetation (Rouse et al., 1974; Tucker, 1979).

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad \text{equation (1)}$$

2.4.2 Normalized Difference Built-up Index (NDBI)

NDBI identifies impervious and built-up areas using spectral contrast between shortwave infrared (SWIR) and near-infrared (NIR) bands. Higher NDBI values indicate urban surfaces such as concrete, asphalt, and rooftops (Y. Zha et al., 2003).

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR} \quad \text{equation (2)}$$

2.4.3 Modified Normalized Difference Water Index (MNDWI)

MNDWI enhances water feature extraction by substituting the near-infrared band with the green band, making it particularly effective in urban environments (Xu, 2006).

$$MNDWI = \frac{Green - SWIR}{Green + SWIR} \quad \text{equation (3)}$$

2.4.4 Land Surface Temperature (LST)

LST was derived from thermal infrared (TIR) bands using the radiative transfer equation (Qin et al., 2001; Weng et al., 2004).

$$LST = \frac{BT}{1 + \left(\frac{\lambda \cdot BT}{\rho} \right) \ln \epsilon} \quad \text{equation (4)}$$

where *BT* is brightness temperature, λ is wavelength, $\rho = h c / \sigma$, and ϵ is surface emissivity estimated via NDVI threshold methods (Sobrino & Jiménez-Muñoz, 2005). LST is essential for urban heat and surface energy balance studies (Voogt & Oke, 2003). All indices and LST were co-registered, resampled to a common grid, and masked for study area boundaries. Radiometric and atmospheric corrections were performed following established protocols (Chander et al., 2009).

2.4.5 Spatial Sampling Framework

A fishnet grid (500×500 m) was generated across the study area, where the centroid (C_{ij}) of each grid cell was extracted and analysed using spatial reference point for LST and its associated predictor variables. The coordinates (C_{ij}) of each grid cell at row i , column j are given by:

$$C_{ij} = \left(X_{\min} + \left(j - \frac{1}{2} \right) * w, Y_{\min} + \left(i - \frac{1}{2} \right) * h \right) \quad \text{equation (5)}$$

where $X_{\min}$, $Y_{\min}$ = coordinates of the study area's lower-left corner; w , h = width and height of the grid cell (500 m each); and i , j = grid cell indices in the row and column directions.

2.4.6 Standard Deviation

The mean, a measure of central tendency, can be used to estimate future observations. This is advantageous because it provides a simple expectation of a subsequent measurement (Ahmed, 2015; Ayeni, 2014; Lee et al., 2015). The most commonly used measure of dispersion is the standard deviation defined by the equation,

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \quad \text{equation (6)}$$

where σ is the sample standard deviation; x_i is the i^{th} observation in the sample; $\bar{x}$ is the sample mean; n is the sample size; and Σ is the summation operator that sums over all observations.

2.4.7 Correlation Matrix

Multi-dimensional stochastic volatility models can be employed to construct correlation matrices by explicitly defining the relationships among variables. In fields such as finance and statistics, understanding these complex interdependencies is essential. To facilitate accurate visualization and interpretation, correlation heatmaps and other graphical correlation plots are widely used. These visualizations leverage the connections between interrelated factors to provide a concise overview of datasets, highlighting patterns, clusters of related variables, and potential outliers through color gradients and intensity variations, defined by the equation,

$$A = a(i, j)_{1 \leq i, j \leq 2n} = \begin{pmatrix} \rho_{(1,1)} & \cdots & \rho_{(1,n)} & \eta_1 & ? \\ \vdots & & \vdots & & \ddots \\ \rho_{(n,1)} & \cdots & \rho_{(n,n)} & ? & \eta_n \\ \eta_1 & & ? & 1 & ? \\ & \ddots & & & \ddots \\ ? & & \eta_n & ? & 1 \end{pmatrix} \quad \text{equation (7)}$$

The completion of a symmetric matrix to a positive definite matrix is possible if its associated graph is chordal and every principal submatrix is positive semidefinite (Grone et al., 1984; Kahl & Günther, 2005). However, the inherent constraints of a correlation matrix, requiring all entries to satisfy $|a_{i,j}| < 1$, introduce an additional layer of complexity (Kahl & Günther, 2005; Laurent, 2001).

2.4.8 Machine Learning Models Development

Machine learning models were implemented to capture both linear and non-linear relationships, with the dataset divided into 80% for training and 20% for testing (Hastie et al., 2009; Kim et al., 2022).

- a) Random Forest (RF) is an ensemble learning method that constructs multiple decision trees and make predictions through majority voting (classification) or averaging (regression):

$$\hat{f}(X) = \frac{1}{B} \sum_{b=1}^B T_b(X) \quad \text{equation (8)}$$

where B = number of trees; and $T_b(X)$ = prediction from the b -th tree. An ensemble of decision trees was trained via bootstrap aggregation (Breiman, 2001). RF handles non-linear relationships and variable interactions effectively.

b) Support Vector Machine (SVM) finds a function $f(X)$ that deviates from observed LST by at most ε :

$$f(X) = \langle w, X \rangle + b \quad \text{equation (9)}$$

$$|LST_i - f(X_i)| \leq \varepsilon \quad \text{equation (10)}$$

Constructs a hyperplane to maximize margin separation (Cortes & Vapnik, 1995).

c) Artificial Neural Network (ANN) are multilayer feedforward network. A layered structure with weighted connections, typically expressed as:

$$LST = f\left(\sum_{j=1}^n w_j * g\left(\sum_{i=1}^m w_{ij} * X_i + b_j\right) + b\right) \quad \text{equation (11)}$$

where $g(\cdot)$ = activation function; w_{ij} , w_j = weights; and b_j , b = bias terms, and f is also an activation function (Rumelhart et al., 1986).

d) Extreme Gradient Boosting (XGBoost) uses additive tree boosting with regularization. An additive ensemble of decision trees minimizing a regularized objective:

$$\hat{y}_i = \sum_{k=1}^K f_k(X_i), \quad f_k \in F \quad \text{equation (12)}$$

where K = number of trees; and f_k = Individual regression tree, $\Omega(f_t)$ controls complexity (T. Chen & Guestrin, 2016).

2.4.9 Shapley Additive Explanations (SHAP)

The SHAP (SHapley Additive exPlanations) framework, grounded in Shapley values from cooperative game theory, offers a consistent and theoretically justified method for interpreting complex machine learning model predictions. The model output for an instance x can be expressed as the sum of a baseline value and the individual feature contributions:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i \quad \text{equation (13)}$$

where $f(x)$ denotes the model prediction, ϕ_0 is the expected model output (baseline), and ϕ_i represents the contribution of the i^{th} feature. The Shapley value for a specific feature i is computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M-|S|-1)!}{M!} |f_x(S \cup \{i\}) - f_x(S)| \quad \text{equation (14)}$$

where N is the full set of input features, S is any subset excluding i , and $f_x(S)$ is the model output for that subset. This additive formulation ensures local accuracy, missingness, and consistency, making SHAP a robust interpretability tool for both linear and non-linear models. A positive ϕ_i indicates that the feature increases the model output relative to the baseline, while a negative value indicates a decrease, allowing for a fair and transparent decomposition of predictions (Lundberg & Lee, 2017; Nohara et al., 2022; Štrumbelj & Kononenko, 2014).

2.4.10 Model Evaluation and Sensitivity Analysis

Three widely adopted evaluation metrics were applied:

$$\text{a) Coefficient of Determination, } R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \quad \text{equation (15)}$$

$$\text{b) Root Mean Squared Error, } RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \quad \text{equation (16)}$$

$$\text{c) Mean Absolute Error, } MAE = \frac{1}{n} \sum |y_i - \hat{y}_i| \quad \text{equation (17)}$$

$$\text{d) Root Mean Squared Logarithmic Error, } RMSLE = \sqrt{\frac{1}{n} \sum (\ln(y_i + 1) - \ln(\hat{y}_i + 1))^2} \quad \text{equation (18)}$$

These indicators provide complementary perspectives: R^2 reflects the proportion of explained variance, RMSE penalizes large deviations, and MAE represents average error magnitude (Willmott & Matsuura, 2005). The Taylor diagram is a graphical tool that summarizes multiple aspects of model performance in a single plot, including the correlation coefficient, standard deviation, and root-mean-square difference (RMSD) between modeled and observed data. It is widely employed in climate science, hydrology, and other geosciences for model evaluation (Taylor, 2001).

The Receiver Operating Characteristic (ROC) curve is widely used to evaluate the performance of classification models. In this study, continuous Land Surface Temperature (LST) values were converted into a binary class $y_i \in \{0,1\}$ using the median value as the threshold (Fawcett, 2006). For each machine learning model, the predicted continuous outputs $\hat{y}_i$ were transformed into probability-like scores p_i , where $0 \leq p_i \leq 1$ (Hastie et al., 2009).

The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) for all possible thresholds:

$$TPR = \frac{TP}{TP + FN} \quad \text{equation (19)}$$

$$FPR = \frac{FP}{FP + TN} \quad \text{equation (20)}$$

where TP, FN, FP, and TN are the numbers of true positives, false negatives, false positives, and true negatives, respectively (Hanley & McNeil, 1982).

The Area Under the Curve (AUC) summarizes model performance:

$$AUC = \int_0^1 TPR(FPR) d(FPR) \quad \text{equation (21)}$$

An AUC of 0.5 indicates random prediction, while a value approaching 1 indicates excellent discriminative ability (Bradley, 1997). ROC curves were generated on the same test dataset to allow direct comparison among models.

2.4.11 UTFVI

The Urban Thermal Field Variance Index (UTFVI) was calculated to assess ecological thermal stress associated with urban heat islands. UTFVI compares the land surface temperature (LST) at each pixel with the mean LST of the study area, following the formulation by:

$$UTFVI = \frac{LST - LST_{mean}}{LST_{mean}} \quad \text{equation (22)}$$

where LST = land surface temperature at each pixel, and LST_{mean} = mean LST of the area of interest. Thermal ecological risk was assessed and classified based on LST across landscapes (D. Zhou et al., 2019). This normalization produces a dimensionless index typically ranging from -1 to 1, where values closer to -1 indicate areas of minimal thermal stress, and values nearer to 1 denote regions experiencing higher heat intensity (Cevik Degerli & Cetin, 2023). UTFVI has been widely adopted in urban climatology studies for mapping urban thermal heterogeneity and identifying heat-prone zones (Weng, 2009).

3 Results

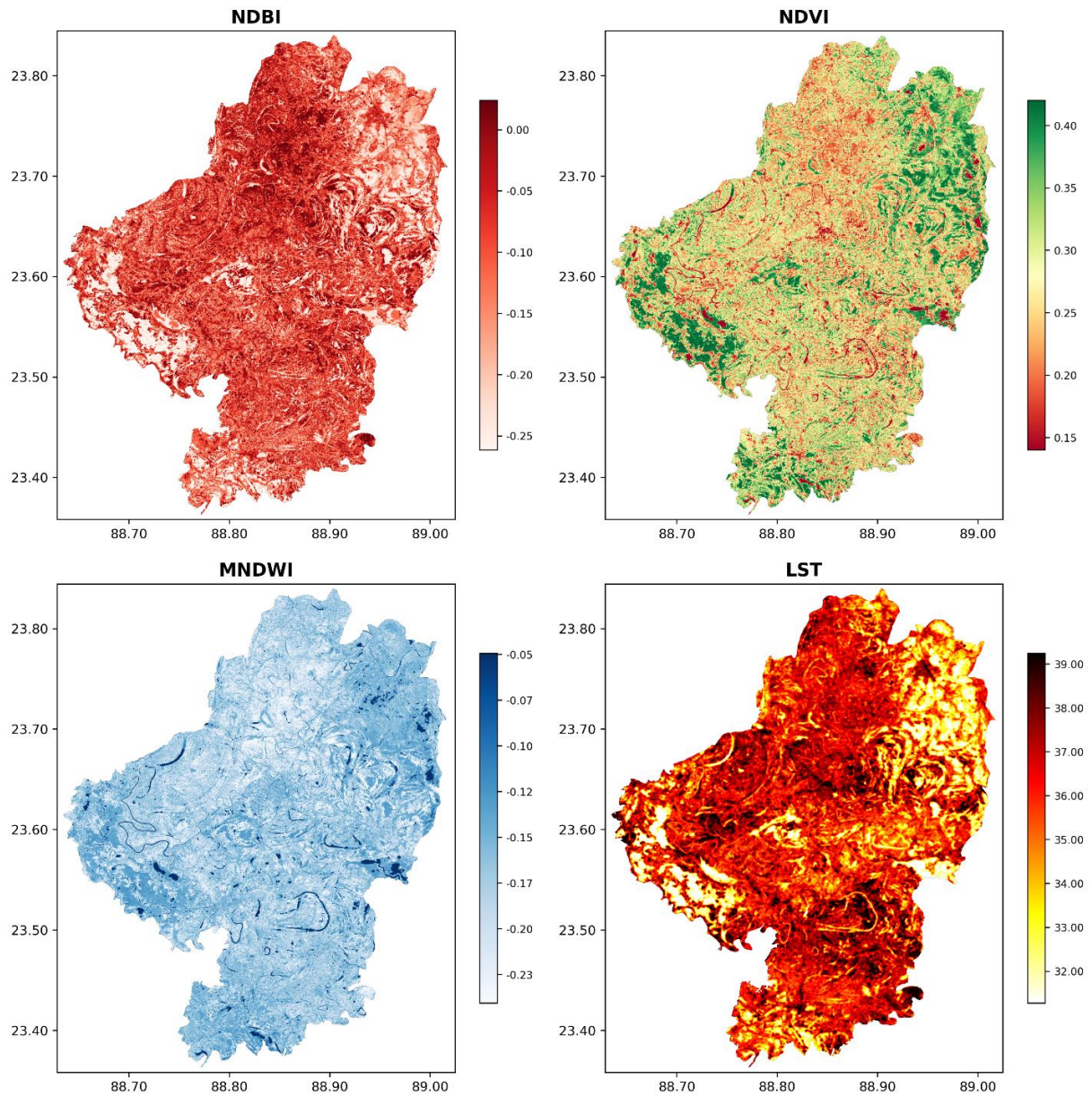


Figure 3: Multi-index map of Chuadanga District.

Figure 3 illustrates the spatial distribution of four important indices in Chuadanga District is described. The map is a combination of the Normalized Difference Built-up Index, the Normalized Difference Vegetation Index, the Modified Normalized Difference Water Index and the Land Surface Temperature to have a descriptive picture of the landscape of the district. NDBI values = -0.26 to 0.02 indicate the presence of built-up areas, NDVI = 0.14 to 0.42 indicates vegetation cover and MNDWI = -0.24 to -0.05 indicates the presence of water. The ranges of LST are 31.27 °C to 39.25 °C and the ranges are used to capture the spatial variation on the surface temperature. All these indices taken together give an combined look at the urbanization, health of the vegetation, accessibility of water in the area, and thermal processes in the district.

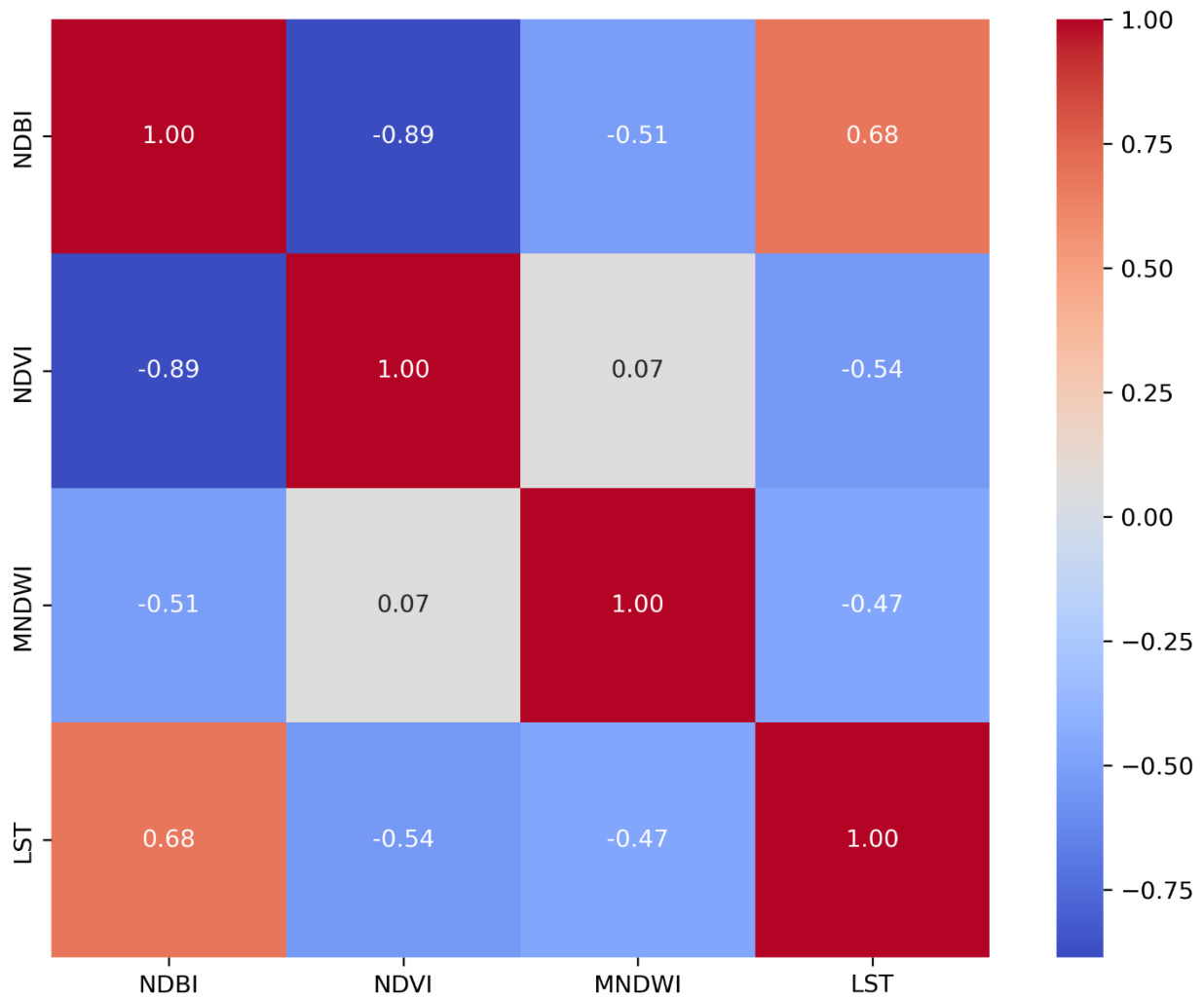


Figure 4: Multi-index heatmap of Chuadanga District.

Figure 4 depicts the complex interactions among urbanization, vegetation, water, and temperature. The negative relationship between NDBI and NDVI (-0.89) states that built-up expansion greatly reduces vegetation cover, and the relationship between NDBI and LST (0.68) is positive which means that built-up surfaces warm up the place. There is a negative relationship between NDVI and LST (-0.54), which indicates that vegetation played a cooling role. MNDWI demonstrates negative moderate relationships with NDBI (-0.51) and LST (-0.47) which show the mitigating effect of water bodies on temperature and its relative lack of frequency in urban areas, yet its weak relationship with NDVI (0.07) indicates that there is no strong interaction between water and the vegetation. Collectively, these patterns underscore the trade-offs between urban growth, ecosystem health, and thermal dynamics.

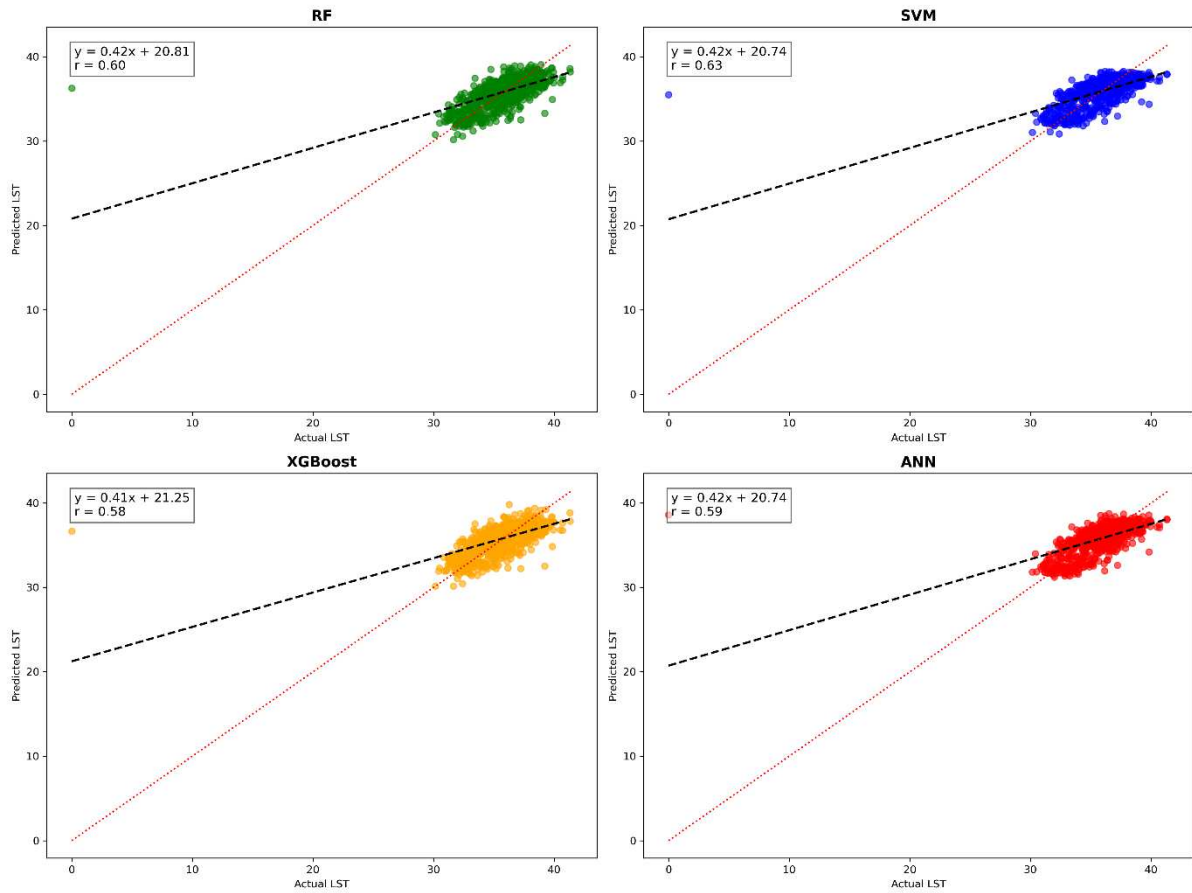


Figure 5: Multiple machine learning approach actual vs predicted LST.

Figure 5 depicts the observed land surface temperature ranged from 30.15 °C to 41.31 °C across the study area. The model predictions exhibited varying ranges: the Random Forest model predicted temperatures between 30.18 °C to 39.10 °C with a correlation of 0.60; Support Vector Machine model predicted 30.86°C to 38.28 °C with the highest correlation of 0.63; XGBoost estimated 30.16 °C to 39.83 °C with a correlation of 0.58; and the Artificial Neural Network predicted 31.04 °C to 39.23 °C with a correlation of 0.59.

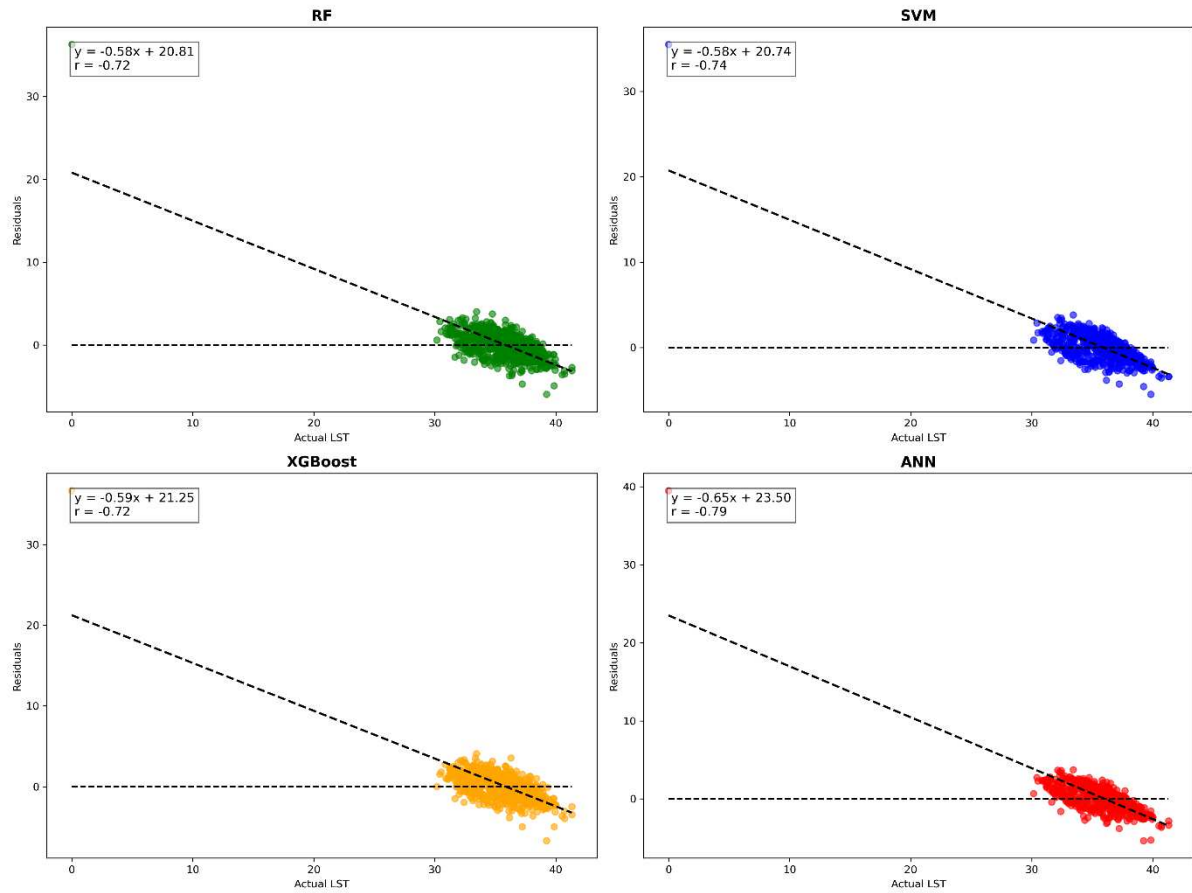


Figure 6: Multiple machine learning approach actual LST vs residuals.

Figure 6 depicts the residuals for the four machine learning models, highlighting the varying degrees of deviation between observed and predicted land surface temperatures. The residuals of the Random Forest model ranged from -5.93°C to 4.02°C , with a correlation of -0.72 . For the Support Vector Machine, residuals ranged from -5.46°C to 3.55°C , exhibiting the highest negative correlation of -0.74 . XGBoost residuals varied between -6.68°C and 4.09°C , with a correlation of -0.72 , while Artificial Neural Network displayed residuals ranging from -5.15°C to 3.70°C , with the strongest negative correlation of -0.78 .

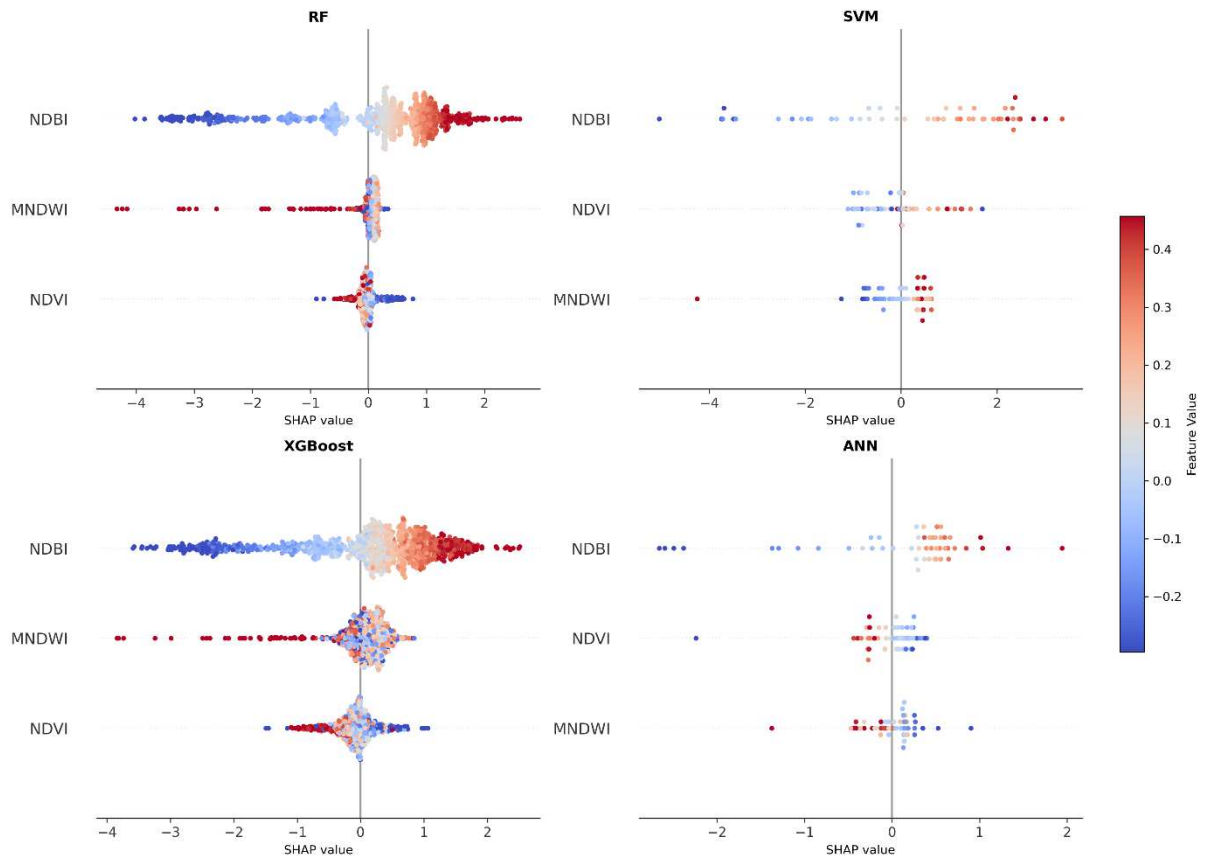


Figure 7: SHAP Beeswarm summary plot showing feature impacts on predictions.

Figure 7 depicts the four machine learning models and the relative influence of their input variables. In the Random Forest model, NDBI exhibited the highest mean absolute SHAP value (1.023), followed by MNDWI (0.148) and NDVI (0.109), indicating that built-up areas were the primary drivers of predictions. In the SVM model, NDBI (1.822) exerted the strongest influence, with NDVI (0.561) and MNDWI (0.492) contributing moderately. For XGBoost, NDBI (0.943) was the most dominant predictor, followed by MNDWI (0.275) and NDVI (0.223). Similarly, in the ANN model, NDBI (0.728) was the most influential, whereas NDVI (0.229) and MNDWI (0.219) had secondary effects. Collectively, these results consistently highlight the dominant role of urbanization, as represented by NDBI, in shaping surface temperature patterns, while vegetation and water bodies exert notable but comparatively smaller influences.

Table II: Multi-Model Evaluation

Model	R ²	RMSE	MAE	RMSLE
RF	0.345191	1.773769	1.075936	0.123896
SVM	0.392975	1.707824	1.024046	0.12272
XGBoost	0.328455	1.796293	1.091552	0.124378
ANN	0.358156	1.756122	1.041381	0.124167

Table II shows the performance evaluation of the machine learning models, which exhibited notable variation across the metrics. The Random Forest model achieved an R² of 0.345, RMSE of 1.77 °C, MAE of 1.08 °C, and RMSLE of 0.124. The Support Vector Machine produced an R² of 0.393, RMSE of 1.71 °C, MAE of 1.02 °C, and RMSLE of 0.123. XGBoost yielded an R² of 0.328, RMSE of 1.80 °C, MAE of 1.09 °C, and RMSLE of 0.124. The Artificial Neural Network resulted in an R² of 0.358, RMSE of 1.76 °C, MAE of 1.04 °C, and RMSLE of

0.124. These results provide a comprehensive overview of the predictive accuracy and error levels of the different models in estimating land surface temperature.

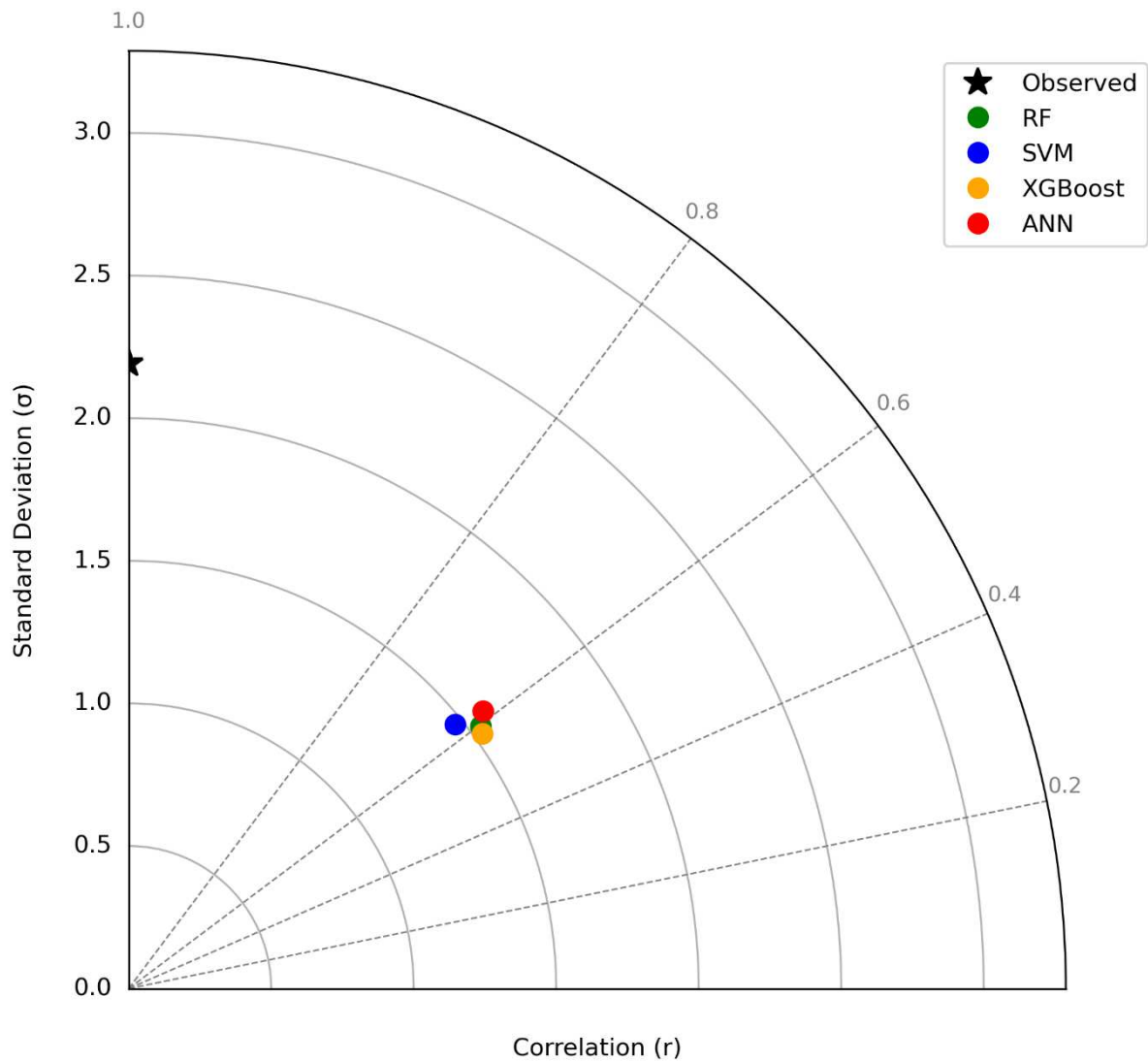


Figure 8: Multiple machine learning approach model performance taylor diagram.

Figure 8 presents the Taylor diagram illustrating the performance of multiple machine learning models in terms of standard deviation and correlation with observed values. The Random Forest model exhibited a standard deviation of 1.54 and a correlation coefficient of 0.60, indicating moderate agreement with observed patterns. The Support Vector Machine achieved a standard deviation of 1.47 and the highest correlation of 0.63, reflecting strong model fidelity. XGBoost demonstrated a standard deviation of 1.53 and a correlation of 0.58, indicating slightly lower consistency compared to the other models. The Artificial Neural Network attained a standard deviation of 1.43 and a correlation of 0.59, highlighting robust predictive performance with moderate variability and satisfactory alignment with observed data.

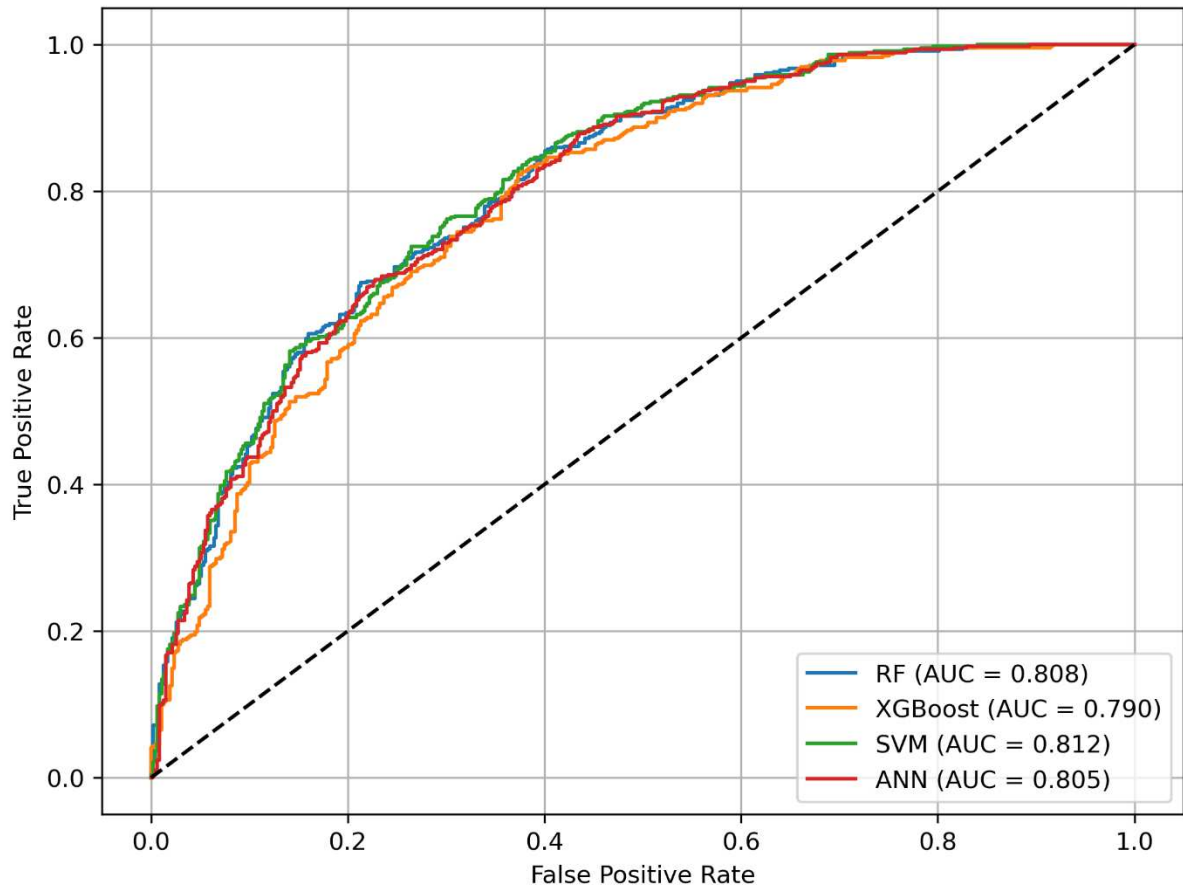


Figure 9: ROC curves for the four machine-learning models.

Figure 9 depicts the ROC curves for the four machine-learning models, showing their ability to distinguish between classes across different thresholds. All models perform above the reference line, indicating strong predictive capability. Among them, SVM achieves the highest discriminative performance with an AUC of 0.8115, followed by RF with 0.8077 and ANN with 0.8045, while XGBoost shows comparatively lower accuracy at 0.7901. Overall, the ROC analysis confirms that all models are effective, with SVM performing best.

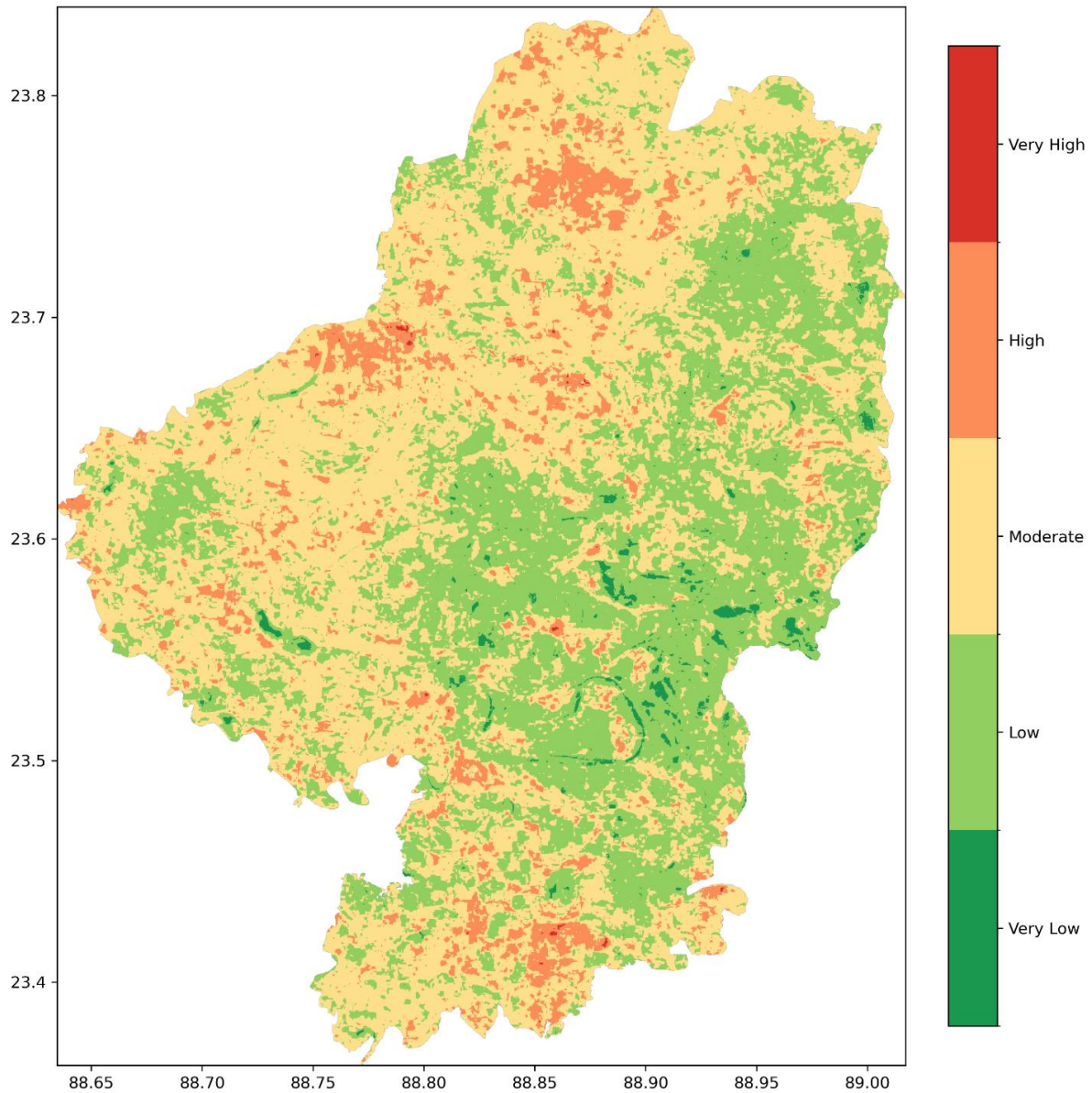


Figure 10: Urban Thermal field Variance of Index of Chuadanga District.

Figure 10 illustrates the Urban Thermal Field Variance Index (UTFVI) for Chuadanga District, categorized into five classes based on the corresponding value ranges. Areas exhibiting very low variance are associated with values between -0.11 and 0.01, whereas low variance spans from 0.01 to 0.13. Moderate variance occurs in regions with values from 0.13 to 0.25, high variance spans between 0.25 and 0.37, and very high variance is observed in areas with values ranging from 0.37 to 0.49. This classification effectively highlights spatial variability of urban thermal behavior across the district.

4 Discussions

The analysis of land surface temperature (LST) and spectral indices in Chuadanga District demonstrates strong interrelationships between urban expansion, vegetation decline, and surface thermal dynamics. Built-up areas, represented by high NDBI values, are closely associated with elevated LST, whereas vegetated and water-covered regions, indicated by higher NDVI and MNDWI values, respectively, exhibit lower surface temperatures. This pattern aligns with previous studies conducted in Bangladesh (Kabir & Ryhan, 2024; Mia et al., 2022) and globally

(Santamouris, 2020), where land cover transformation has been recognized as a principal driver of the Urban Heat Island (UHI) effect.

The relationships among the indices indicate that urbanization drives the replacement of vegetated areas with impervious surfaces, thereby reducing evapotranspiration and increasing the heat storage capacity of the landscape. A strong negative correlation between NDBI and NDVI underscores the trade-off between built-up expansion and vegetation coverage, whereas a positive association between NDBI and LST highlights the warming effects of urban materials such as concrete and asphalt. The moderate negative relationship between MNDWI and LST suggests that water bodies contribute to localized cooling, although their limited presence reduces their overall influence. Collectively, these relationships emphasize the importance of conserving ecological surfaces like vegetation and water bodies, to mitigate thermal stress in urban environments.

Among the machine learning models applied for LST estimation, the Support Vector Machine exhibited the highest predictive accuracy and robustness, reflecting its ability to capture non-linear relationships between spectral indices and surface temperature. The Random Forest, XGBoost, and Artificial Neural Network models also provided reliable results, demonstrating the effectiveness of ensemble and neural network approaches in urban thermal mapping. The SHAP-based feature importance analysis identified NDBI as the dominant predictor across all models, reaffirming the strong influence of urbanization on thermal variability, while NDVI and MNDWI had moderate yet meaningful contributions.

Based on the effects of built-up areas, the spatial analysis of thermal variability indicated that high and very high thermal zones are predominantly concentrated in central urban clusters and industrial areas. This spatial distribution reflects a pattern of early-stage urban heat amplification, which is a characteristic of small but rapidly developing cities in Bangladesh. Similar phenomena have been observed in Dhaka, Khulna, and Rajshahi, where unplanned urban expansion and reduced green cover have exacerbated urban heat risks (M. N. Rahman et al., 2022; Salam et al., 2024).

From a policy perspective, these findings underscore the urgent need to integrate LST-based indicators into the urban climate adaptation strategies of Bangladesh. While international frameworks such as the Paris Agreement (UNFCCC, 2015) and the Sustainable Development Goals (SDGs 11 and 13) promote urban resilience and heat mitigation, national instruments – including the Bangladesh Climate Change Strategy and Action Plan (BCCSAP 2009) and the National Adaptation Plan (2022–2050) have yet to operationalize satellite-based thermal assessments. Integrating LST monitoring within Detailed Area Plans (DAPs) and Urban Master Plans could improve zoning, heat-risk identification, and the implementation of nature-based cooling interventions, such as urban forestry, reflective materials, and green roofs (Mia et al., 2022; UNDESA, 2023). Moreover, strengthening collaboration among the Department of Environment (DoE), Bangladesh Meteorological Department (BMD), and Bangladesh Space Research and Remote Sensing Organization (SPARRSO) can further facilitate a coordinated national framework for LST-based urban climate management.

5 Limitations

- i. **Limitations in spatial and temporal resolution:** Medium resolution (Landsat 30 m) imagery might not provide sufficient information to detect the presence of micro-urban variations and fine scale heterogeneity in surface temperature (Weng, 2009).
- ii. **Effects of clouds and the atmosphere:** The optical satellite data is usually compromised by cloud cover, haze and atmospheric water vapor effects that worsen the quality of reflectance, and decrease model accuracy (Sobrino et al., 2004).
- iii. **Uncertainty of surface emissivity:** The emissivity of the different land covers can vary and as such, this can introduce error in LST retrieval, especially at the boundary or transitional zone (Jiménez-Muñoz & Sobrino, 2008).
- iv. **Spatial correlation and scale disparity:** Dissimilarity at spatial scales of predictor indexes (NDVI, NDBI, MNDWI) and LST can cause assumptions and autocorrelation of the machine learning outcomes (Hu & Weng, 2009).

- v. **Dependency on optical data:** Use of the optical sensors constrains the availability of data in the cases of cloudiness, or monsoon season, combining thermal or microwave data might alleviate this constraint (Weng, 2009).

Conclusion

The study highlights the massive effect of urbanization, vegetation and water bodies in land surface temperatures within Chuadanga District. High NDBI values, which pointed at built-up areas, were linked to high LST, but vegetation and water bodies minimized thermal stress. The Support Vector Machine (SVM) was the most successful model out of all the machine learning models that addressed non-linear relationships and identified NDBI as the most important model in defining the surface temperature, SHapley Additive exPlanations (SHAP). The thermal stress concentration on the cores of urban cities and the industrial zones were found to be at high levels as shown by Urban Thermal Field Variance Index (UTFVI), which depicts a beginning of urban heat enhancement. These results highlight the importance of integrating the LST-based measures of urban planning and climate adaptation measures, such as zoning, heat-risk mapping, and nature-based cooling measures, i.e. the use of green roofs and urban forestry. In general, the combination of remote sensing, spectral indices, and machine learning offers a powerful, data-driven model of the evaluation of urban thermal dynamics and promotes climate-resilient planning of new secondary cities in Bangladesh.

CRedit authorship contribution statement

Golam Rasul: Conceptualization, Literature review, Methodology development, Data Analysis, Visualization, Writing – original draft, review & editing.; **Md. Akik Tanjil Jihan:** Conceptualization, Methodology development, Data Analysis, Visualization, Writing – original draft, review & editing.; **Md Sohan Khan:** Literature review, Methodology development, Visualization, Writing – original draft, review & editing.; **Naeem Sheikh:** Literature review, Writing – original draft, review & editing.; **Samsunnahar Popy:** Supervision, Writing – review & editing.

Data availability statement

All data generated or analyzed during this study are included in this published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Reference

- Abd-Elmabod, S. K., Jiménez-González, M. A., Jordán, A., Zhang, Z., Mohamed, E. S., Hammam, A. A., El Baroudy, A. A., Abdel-Fattah, M. K., Abdelfattah, M. A., & Jones, L. (2022). Past and future impacts of

- urbanisation on land surface temperature in Greater Cairo over a 45 year period. *Egyptian Journal of Remote Sensing and Space Science*, 25(4), 961–974. <https://doi.org/10.1016/j.ejrs.2022.10.001>
- Ahmed, S. N. (2015). Essential statistics for data analysis. In *Physics and Engineering of Radiation Detection* (pp. 541–593). Elsevier. <https://doi.org/10.1016/B978-0-12-801363-2.00009-7>
- Alexander, C. (2020). Normalised difference spectral indices and urban land cover as indicators of land surface temperature (LST). *International Journal of Applied Earth Observation and Geoinformation*, 86. <https://doi.org/10.1016/j.jag.2019.102013>
- Avdan, U., & Jovanovska, G. (2016). Algorithm for automated mapping of land surface temperature using LANDSAT 8 satellite data. *Journal of Sensors*, 2016. <https://doi.org/10.1155/2016/1480307>
- Ayeni, A. W. (2014). *Empirics of standard deviation*. <https://doi.org/10.13140/2.1.1444.6729>
- Bradley, A. P. (1997). The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition*, 30(7), 1145–1159. [https://doi.org/10.1016/S0031-3203\(96\)00142-2](https://doi.org/10.1016/S0031-3203(96)00142-2)
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/https://link.springer.com/article/10.1023/A%3A1010933404324>
- Cetin, M., Ozenen Kavlak, M., Senyel Kurkcuoglu, M. A., Bilge Ozturk, G., Cabuk, S. N., & Cabuk, A. (2024). Determination of land surface temperature and urban heat island effects with remote sensing capabilities: the case of Kayseri, Türkiye. *Natural Hazards*, 120(6), 5509–5536. <https://doi.org/10.1007/s11069-024-06431-5>
- Cevik Degerli, B., & Cetin, M. (2023). Evaluation of UTFVI index effect on climate change in terms of urbanization. *Environmental Science and Pollution Research*, 30(30), 75273–75280. <https://doi.org/10.1007/s11356-023-27613-x>
- Chander, G., Markham, B. L., & Helder, D. L. (2009). Summary of current radiometric calibration coefficients for Landsat MSS, TM, ETM+, and EO-1 ALI sensors. *Remote Sensing of Environment*, 113(5), 893–903. <https://doi.org/10.1016/j.rse.2009.01.007>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 13-17-August-2016*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Chen, X. L., Zhao, H. M., Li, P. X., & Yin, Z. Y. (2006). Remote sensing image-based analysis of the relationship between urban heat island and land use/cover changes. *Remote Sensing of Environment*, 104(2), 133–146. <https://doi.org/10.1016/j.rse.2005.11.016>
- Choe, Y. J., & Yom, J. H. (2020). Improving accuracy of land surface temperature prediction model based on deep-learning. *Spatial Information Research*, 28(3), 377–382. <https://doi.org/10.1007/s41324-019-00299-5>
- Cortes, C., & Vapnik, V. (1995). Support-Vector Networks Editor. *Machine Learning*, 20, 273–297. <https://doi.org/https://link.springer.com/article/10.1007/BF00994018>
- Farid, N., Moazzam, M. F. U., Ahmad, S. R., Coluzzi, R., & Lanfredi, M. (2022). Monitoring the Impact of Rapid Urbanization on Land Surface Temperature and Assessment of Surface Urban Heat Island Using Landsat in Megacity (Lahore) of Pakistan. *Frontiers in Remote Sensing*, 3. <https://doi.org/10.3389/frsen.2022.897397>
- Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874. <https://doi.org/10.1016/j.patrec.2005.10.010>

- Ferrelli, F., Huamantino Cisneros, M. A., Delgado, A. L., & Piccolo, M. C. (2018). Spatial and temporal analysis of the LST-NDVI relationship for the study of land cover changes and their contribution to urban planning in monte hermoso, Argentina. *Documents d'Analisi Geografica*, 64(1), 25–47. <https://doi.org/10.5565/rev/dag.355>
- Fleck, R., Gill, R. L., Saadeh, S., Pettit, T., Wooster, E., Torpy, F., & Irga, P. (2022). Urban green roofs to manage rooftop microclimates: A case study from Sydney, Australia. *Building and Environment*, 209, 108673. <https://doi.org/10.1016/J.BUILDENV.2021.108673>
- Foga, S., Scaramuzza, P. L., Guo, S., Zhu, Z., Dilley, R. D., Beckmann, T., Schmidt, G. L., Dwyer, J. L., Joseph Hughes, M., & Laue, B. (2017). Cloud detection algorithm comparison and validation for operational Landsat data products. *Remote Sensing of Environment*, 194, 379–390. <https://doi.org/10.1016/j.rse.2017.03.026>
- Gao, P., Du, W., Lei, Q., Li, J., Zhang, S., & Li, N. (2023). NDVI forecasting model based on the combination of time series decomposition and CNN–LSTM. *Water Resources Management*, 37(4), 1481–1497. <https://doi.org/10.1007/s11269-022-03419-3>
- Ghosh, S., Chatterjee, N. Das, & Dinda, S. (2019). Relation between urban biophysical composition and dynamics of land surface temperature in the Kolkata metropolitan area: a GIS and statistical based analysis for sustainable planning. *Modeling Earth Systems and Environment*, 5(1), 307–329. <https://doi.org/10.1007/s40808-018-0535-9>
- Guha, S., Govil, H., & Diwan, P. (2020). Monitoring LST-NDVI relationship using premonsoon Landsat datasets. *Advances in Meteorology*, 2020. <https://doi.org/10.1155/2020/4539684>
- Halim, M. A., Chowdhury, M. S. H., Muhammed, N., Rahman, M., & Koike, M. (2008). Sap production from khejur palm (phoenix sylvestris roxb) husbandry: A substantial means of seasonal livelihood in rural bangladesh. *Forests Trees and Livelihoods*, 18(3), 305–318. <https://doi.org/10.1080/14728028.2008.9752638>
- Hanley, J. A., & McNeil, B. J. (1982). The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1), 29–36. <https://doi.org/10.1148/radiology.143.1.7063747>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer New York. <https://doi.org/10.1007/978-0-387-84858-7>
- Hovorushchenko, T., & Alekseiko, V. (2024). Land surface temperature forecasting in the context of the development of sustainable cities and communities. *Computer Systems and Information Technologies*, 3, 6–12. <https://doi.org/10.31891/csit-2024-3-1>
- Hovorushchenko, T., Alekseiko, V., & Levashenko, V. (2024, December 5). Machine learning methods' comparison for land surface temperatures forecasting due to climate classification. *1st International Workshop on Advanced Applied Information Technologies*. <https://ceur-ws.org/Vol-3899/paper6.pdf>
- Howard, L. (2012). *The Climate of London*. Cambridge University Press. <https://doi.org/10.1017/CBO9781139226899>
- Hu, X., & Weng, Q. (2009). Estimating impervious surfaces from medium spatial resolution imagery using the self-organizing map and multi-layer perceptron neural networks. *Remote Sensing of Environment*, 113(10), 2089–2102. <https://doi.org/10.1016/j.rse.2009.05.014>
- IBM. (2025). *What is Machine Learning?* <https://www.ibm.com/think/topics/machine-learning>

- Islam, Md. N., Haque, Md. I., & Hossain, Md. A. (2023). Spatial pattern of arsenic concentration and associated noncarcinogenic health risk assessment: a case study on Gangni Union of Chuadanga district of Bangladesh. *Environmental Systems Research*, 12(1), 29. <https://doi.org/10.1186/s40068-023-00313-8>
- Jiang, N., Crooks, A., & Crooks, A. T. (2020). *UTILIZING AGENTS TO EXPLORE URBAN SHRINKAGE: A CASE STUDY OF DETROIT*. <https://www.researchgate.net/publication/341490879>
- Jiménez-Muñoz, J. C., & Sobrino, J. A. (2008). Split-window coefficients for land surface temperature retrieval from low-resolution thermal infrared sensors. *IEEE Geoscience and Remote Sensing Letters*, 5(4), 806–809. <https://doi.org/10.1109/LGRS.2008.2001636>
- Julien, Y., Sobrino, J. A., & Verhoef, W. (2006). Changes in land surface temperatures and NDVI values over Europe between 1982 and 1999. *Remote Sensing of Environment*, 103(1), 43–55. <https://doi.org/10.1016/j.rse.2006.03.011>
- Kabir, M. H., & Ryhan, M. A. (2024). Spatiotemporal variation of land surface temperature and urban heat islands in Dhaka City: a geospatial approach. *The Dhaka University Journal of Earth and Environmental Sciences*, 13(1), 1–19. <https://doi.org/10.3329/dujees.v13i1.77553>
- Kahl, C., & Günther, M. (2005). Complete the correlation matrix. In *From Nano to Space* (pp. 229–244). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-74238-8_17
- Kaufmann, R. K., Zhou, L., Myneni, R. B., Tucker, C. J., Slayback, D., Shabanov, N. V., & Pinzon, J. (2003). The effect of vegetation on surface temperature: A statistical analysis of NDVI and climate data. *Geophysical Research Letters*, 30(22). <https://doi.org/10.1029/2003GL018251>
- Kim, M., Kim, D., & Kim, G. (2022). Examining the relationship between land use/land cover (LULC) and land surface temperature (LST) using explainable artificial intelligence (XAI) models: a case study of Seoul, South Korea. *International Journal of Environmental Research and Public Health*, 19(23). <https://doi.org/10.3390/ijerph192315926>
- Krasniqi, V., & Rapuca, A. (2024). Impact Assessment of Urban Development Patterns on Land Surface Temperature and Urban Heat Islands Using Remote Sensing Techniques – A Case Study of Prishtina, Kosovo. *Journal of Ecological Engineering*, 25(10), 91–100. <https://doi.org/10.12911/22998993/191784>
- Laurent, M. (2001). Matrix completion problems. In *Encyclopedia of Optimization* (pp. 1311–1319). Springer US. https://doi.org/10.1007/0-306-48332-7_271
- Lee, D. K., In, J., & Lee, S. (2015). Standard deviation and standard error of the mean. *Korean Journal of Anesthesiology*, 68(3), 220. <https://doi.org/10.4097/kjae.2015.68.3.220>
- Li, L., Zhou, X., Chen, L., Chen, L., Zhang, Y., & Liu, Y. (2020). Estimating urban vegetation biomass from sentinel-2A image data. *Forests*, 11(2). <https://doi.org/10.3390/f11020125>
- Li, Q., & Zheng, H. (2023). Prediction of summer daytime land surface temperature in urban environments based on machine learning. *Sustainable Cities and Society*, 97, 104732. <https://doi.org/10.1016/j.scs.2023.104732>
- Li, S., Zhao, Z., Miaomiao, X., & Wang, Y. (2010). Investigating spatial non-stationary and scale-dependent relationships between urban surface temperature and environmental factors using geographically weighted regression. *Environmental Modelling and Software*, 25(12), 1789–1800. <https://doi.org/10.1016/j.envsoft.2010.06.011>

- Li, Z. L., Tang, B. H., Wu, H., Ren, H., Yan, G., Wan, Z., Trigo, I. F., & Sobrino, J. A. (2013). Satellite-derived land surface temperature: Current status and perspectives. In *Remote Sensing of Environment* (Vol. 131, pp. 14–37). <https://doi.org/10.1016/j.rse.2012.12.008>
- Liang, M., Zhang, L., Wu, S., Zhu, Y., Dai, Z., Wang, Y., Qi, J., Chen, Y., & Du, Z. (2023). A high-resolution land surface temperature downscaling method based on geographically weighted neural network regression. *Remote Sensing*, 15(7). <https://doi.org/10.3390/rs15071740>
- Lu, B., Charlton, M., Harris, P., & Fotheringham, A. S. (2014). Geographically weighted regression with a non-Euclidean distance metric: A case study using hedonic house price data. *International Journal of Geographical Information Science*, 28(4), 660–681. <https://doi.org/10.1080/13658816.2013.865739>
- Lundberg, S., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *31st Conference on Neural Information Processing Systems*, 4766–4777. <https://doi.org/https://doi.org/10.48550/arXiv.1705.07874>
- Macarof, P., & Statescu, F. (2017). Comparison of NDBI and NDVI as indicators of surface urban heat island effect in Landsat 8 imagery: a case study of Iasi. *Present Environment and Sustainable Development*, 11(2), 141–150. <https://doi.org/10.1515/pesd-2017-0032>
- Marando, F., Heris, M. P., Zulian, G., Udías, A., Mentaschi, L., Chrysoulakis, N., Parastatidis, D., & Maes, J. (2022). Urban heat island mitigation by green infrastructure in European Functional Urban Areas. *Sustainable Cities and Society*, 77. <https://doi.org/10.1016/j.scs.2021.103564>
- Masson, V., Lemonsu, A., Hidalgo, J., & Voogt, J. (2020). Urban Climates and Climate Change. *Annual Review of Environment and Resources*, 45(1), 411–444. <https://doi.org/10.1146/annurev-environ-012320-083623>
- Meng, Q., Liu, W., Zhang, L., Allam, M., Bi, Y., Hu, X., Gao, J., Hu, D., & Jancsó, T. (2022). Relationships between land surface temperatures and neighboring environment in highly urbanized areas: seasonal and scale effects analyses of Beijing, China. *Remote Sensing*, 14(17). <https://doi.org/10.3390/rs14174340>
- Mentaschi, L., Duveiller, G., Zulian, G., Corbane, C., Pesaresi, M., Maes, J., Stocchino, A., & Feyen, L. (2022). Global long-term mapping of surface temperature shows intensified intra-city urban heat island extremes. *Global Environmental Change*, 72. <https://doi.org/10.1016/j.gloenvcha.2021.102441>
- Mia, M. B., Hasan, M. A., Gazi, M. Y., Chowdhury, M., & Montakim, N. Al. (2022). Impact of urbanization on land use-land cover, land surface temperature and urban heat islands using multispectral satellite images: an implication in the district towns of northwestern Bangladesh. *The Dhaka University Journal of Earth and Environmental Sciences*, 17–28. <https://doi.org/10.3329/dujees.v10i3.59068>
- Morsy, S., & Hadi, M. (2022). Impact of land use/land cover on land surface temperature and its relationship with spectral indices in Dakahlia Governorate, Egypt. *International Journal of Engineering and Geosciences*, 7(3), 272–282. <https://doi.org/10.26833/ijeg.978961>
- Mustafa, E. K., Co, Y., Liu, G., Kaloop, M. R., Beshr, A. A., Zarzoura, F., & Sadek, M. (2020). Study for Predicting Land Surface Temperature (LST) Using Landsat Data: A Comparison of Four Algorithms. *Advances in Civil Engineering*, 2020. <https://doi.org/10.1155/2020/7363546>
- Nations, U., of Economic, D., Affairs, S., & Division, P. (2018). *World Urbanization Prospects The 2018 Revision*. <https://population.un.org/wup/assets/WUP2018-Report.pdf>
- Nohara, Y., Matsumoto, K., Soejima, H., & Nakashima, N. (2022). Explanation of machine learning models using shapley additive explanation and application for real data in hospital. *Computer Methods and Programs in Biomedicine*, 214, 106584. <https://doi.org/10.1016/j.cmpb.2021.106584>

- Ogashawara, I., & Bastos, V. (2012). A Quantitative Approach for Analyzing the Relationship between Urban Heat Islands and Land Cover. *Remote Sensing*, 4(11), 3596–3618. <https://doi.org/10.3390/rs4113596>
- Oke, T. R. (1982). The energetic basis of the urban heat island. In *Quart. J. R. Met. Soc* (Vol. 108, Issue 455).
- Perkins-Kirkpatrick, S. E., & Lewis, S. C. (2020). Increasing trends in regional heatwaves. *Nature Communications*, 11(1). <https://doi.org/10.1038/s41467-020-16970-7>
- Qin, Z., Karnieli, A., & Berliner, P. (2001). A mono-window algorithm for retrieving land surface temperature from Landsat TM data and its application to the Israel-Egypt border region. *International Journal of Remote Sensing*, 22(18), 3719–3746. <https://doi.org/10.1080/01431160010006971>
- Rahman, M. A., Afridi, S., Hossain, M. B., Rana, M., Al Masum, A., Rahman, M. M., & Al-Maruf, A. (2024). Nexus between heat wave, food security and human health (HFH): Developing a framework for livelihood resilience in Bangladesh. *Environmental Challenges*, 14. <https://doi.org/10.1016/j.envc.2023.100802>
- Rahman, M. N., Rony, M. R. H., Jannat, F. A., Pal, S. C., Islam, M. S., Alam, E., & Islam, A. R. M. T. (2022). Impact of urbanization on urban heat island intensity in major districts of Bangladesh using remote sensing and geo-spatial tools. *Climate*, 10(1). <https://doi.org/10.3390/cli10010003>
- Rashid, N., Alam, J. A. M. M., Chowdhury, M. A., & Islam, S. L. U. (2022). Impact of landuse change and urbanization on urban heat island effect in Narayanganj city, Bangladesh: A remote sensing-based estimation. *Environmental Challenges*, 8. <https://doi.org/10.1016/j.envc.2022.100571>
- Rasool, R., Fayaz, A., Shafiq, M. ul, Singh, H., & Ahmed, P. (2021). Land use land cover change in Kashmir Himalaya: Linking remote sensing with an indicator based DPSIR approach. *Ecological Indicators*, 125. <https://doi.org/10.1016/j.ecolind.2021.107447>
- Robbiati, F. O., Cáceres, N., Hick, E. C., Suarez, M., Soto, S., Barea, G., Matoff, E., Galetto, L., & Imhof, L. (2022). Vegetative and thermal performance of an extensive vegetated roof located in the urban heat island of a semiarid region. *Building and Environment*, 212, 108791. <https://doi.org/10.1016/J.BUILDENV.2022.108791>
- Rouse, J. W. , Jr., Haas, R. H., Schell, J. A., & Deering, D. W. (1974). Monitoring vegetation systems in the Great Plains with ERTS. *NASA SP-351*, 309–317. <https://ntrs.nasa.gov/api/citations/19740022614/downloads/19740022614.pdf>
- Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533–536. <https://doi.org/10.1038/323533a0>
- Salam, M., Islam, M. K., Jahan, I., & Chowdhury, M. A. (2024). Assessing the impacts of vegetation loss and land surface temperature on Surface Urban Heat Island (SUHI) in Gazipur District, Bangladesh. *Computational Urban Science*, 4(1). <https://doi.org/10.1007/s43762-024-00136-y>
- Sameh, S., Zarzoura, F., & El-Mewafi, M. (2022). Automated mapping of urban heat island to predict land surface temperature and land use/cover change using machine learning algorithms: Mansoura City. *International Journal of Geoinformatics*, 18(6), 47–67. <https://doi.org/10.52939/ijg.v18i6.2461>
- Santamouris, M. (2020). Recent progress on urban overheating and heat island research. Integrated assessment of the energy, environmental, vulnerability and health impact. Synergies with the global climate change. In *Energy and Buildings* (Vol. 207). Elsevier Ltd. <https://doi.org/10.1016/j.enbuild.2019.109482>
- Sobremonte-Maglipon, P. A., Olfato-Parojinog, A., Almadrones-Reyes, K. J., Limbo-Dizon, J. E., & Dagamac, N. H. A. (2024). Tracking the temporal changes in land surface temperature, vegetation, and built-up patterns

- in Rizal Province, Philippines using landsat imagery. *Geoplanning*, 11(1), 71–84. <https://doi.org/10.14710/geoplanning.11.1.71-84>
- Sobrino, J. A., & Jiménez-Muñoz, J. C. (2005). Land surface temperature retrieval from thermal infrared data: An assessment in the context of the Surface Processes and Ecosystem Changes Through Response Analysis (SPECTRA) mission. *Journal of Geophysical Research D: Atmospheres*, 110(16), 1–10. <https://doi.org/10.1029/2004JD005588>
- Sobrino, J. A., Jiménez-Muñoz, J. C., & Paolini, L. (2004). Land surface temperature retrieval from LANDSAT TM 5. *Remote Sensing of Environment*, 90(4), 434–440. <https://doi.org/10.1016/j.rse.2004.02.003>
- Song, X. P. (2018). Global Estimates of Ecosystem Service Value and Change: Taking Into Account Uncertainties in Satellite-based Land Cover Data. *Ecological Economics*, 143, 227–235. <https://doi.org/10.1016/j.ecolecon.2017.07.019>
- Steven, R., Pickering, C., & Guy Castley, J. (2011). A review of the impacts of nature based recreation on birds. In *Journal of Environmental Management* (Vol. 92, Issue 10, pp. 2287–2294). Academic Press. <https://doi.org/10.1016/j.jenvman.2011.05.005>
- Štrumbelj, E., & Kononenko, I. (2014). Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems*, 41(3), 647–665. <https://doi.org/10.1007/s10115-013-0679-x>
- Taylor, K. E. (2001). Summarizing multiple aspects of model performance in a single diagram. *Journal of Geophysical Research Atmospheres*, 106(D7), 7183–7192. <https://doi.org/10.1029/2000JD900719>
- Tomlinson, C. J., Chapman, L., Thornes, J. E., & Baker, C. (2011). Remote sensing land surface temperature for meteorology and climatology: A review. In *Meteorological Applications* (Vol. 18, Issue 3, pp. 296–306). John Wiley and Sons Ltd. <https://doi.org/10.1002/met.287>
- Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2), 127–150. [https://doi.org/10.1016/0034-4257\(79\)90013-0](https://doi.org/10.1016/0034-4257(79)90013-0)
- Uisso, A. M., & Tanrıvermiş, H. (2021). Driving factors and assessment of changes in the use of arable land in Tanzania. *Land Use Policy*, 104. <https://doi.org/10.1016/j.landusepol.2021.105359>
- UNDESA. (2023). *Towards Sustainable Development For All Department of Economic and Social Affairs*. <https://www.un.org/en/desa/highlights-report-2022-2023>.
- UNFCCC. (2015). *Adoption of the Paris Agreement*. <https://unfccc.int/resource/docs/2015/cop21/eng/109r01.pdf>
- Viana, C. M., Oliveira, S., Oliveira, S. C., & Rocha, J. (2019). Land Use/Land Cover Change Detection and Urban Sprawl Analysis. In *Spatial Modeling in GIS and R for Earth and Environmental Sciences* (pp. 621–651). Elsevier. <https://doi.org/10.1016/b978-0-12-815226-3.00029-6>
- Voogt, J. A., & Oke, T. R. (2003). Thermal remote sensing of urban climates. *Remote Sensing of Environment*, 86(3), 370–384. [https://doi.org/10.1016/S0034-4257\(03\)00079-8](https://doi.org/10.1016/S0034-4257(03)00079-8)
- Wang, H., Zhang, Y., Tsou, J. Y., & Li, Y. (2017). Surface urban heat island analysis of shanghai (China) based on the change of land use and land cover. *Sustainability (Switzerland)*, 9(9). <https://doi.org/10.3390/su9091538>
- Weng, Q. (2009). Thermal infrared remote sensing for urban climate and environmental studies: Methods, applications, and trends. In *ISPRS Journal of Photogrammetry and Remote Sensing* (Vol. 64, Issue 4, pp. 335–344). <https://doi.org/10.1016/j.isprsjprs.2009.03.007>

- Weng, Q., Lu, D., & Schubring, J. (2004). Estimation of land surface temperature-vegetation abundance relationship for urban heat island studies. *Remote Sensing of Environment*, 89(4), 467–483. <https://doi.org/10.1016/j.rse.2003.11.005>
- Willmott, C. J., & Matsuura, K. (2005). Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Climate Research*, 30(1), 79–82. <https://doi.org/10.3354/cr030079>
- Xia, H., Chen, Y., Zhao, Y., & Chen, Z. (2018). “Regression-then-fusion” or “fusion-then-regression”? A theoretical analysis for generating high spatiotemporal resolution land surface temperatures. *Remote Sensing*, 10(9). <https://doi.org/10.3390/rs10091382>
- Xiang, Y., Tang, Y., Wang, Z., Peng, C., Huang, C., Dian, Y., Teng, M., & Zhou, Z. (2023). Seasonal variations of the relationship between spectral indexes and land surface temperature based on local climate zones: a study in three Yangtze River megacities. *Remote Sensing*, 15(4), 870. <https://doi.org/10.3390/rs15040870>
- Xu, H. (2006). Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*, 27(14), 3025–3033. <https://doi.org/10.1080/01431160600589179>
- Yuan, J., Lv, X., Dou, F., & Yao, J. (2019). Change analysis in urban areas based on statistical features and temporal clustering using TerraSAR-X time-series images. *Remote Sensing*, 11(8). <https://doi.org/10.3390/rs11080957>
- Zha, F., Lu, L., Wang, R., Zhang, S., Cao, S., Baqa, M. F., Li, Q., & Chen, F. (2024). Understanding fine-scale heat health risks and the role of green infrastructure based on remote sensing and socioeconomic data in the megacity of Beijing, China. *Ecological Indicators*, 160, 111847. <https://doi.org/10.1016/J.ECOLIND.2024.111847>
- Zha, Y., Gao, J., & Ni, S. (2003). Use of normalized difference built-up index in automatically mapping urban areas from TM imagery. *International Journal of Remote Sensing*, 24(3), 583–594. <https://doi.org/10.1080/01431160304987>
- Zhang, L., Gove, J. H., & Heath, L. S. (2005). Spatial residual analysis of six modeling techniques. *Ecological Modelling*, 186(2), 154–177. <https://doi.org/10.1016/j.ecolmodel.2005.01.007>
- Zhou, D., Xiao, J., Bonafoni, S., Berger, C., Deilami, K., Zhou, Y., Froking, S., Yao, R., Qiao, Z., & Sobrino, J. A. (2019). Satellite remote sensing of surface urban heat islands: Progress, challenges, and perspectives. In *Remote Sensing* (Vol. 11, Issue 1). MDPI AG. <https://doi.org/10.3390/rs11010048>
- Zhou, X., & Chen, H. (2018). Impact of urbanization-related land use land cover changes and urban morphology changes on the urban heat island phenomenon. *Science of the Total Environment*, 635, 1467–1476. <https://doi.org/10.1016/j.scitotenv.2018.04.091>